# Tree Implementations

Chapter 16

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# Nodes in a Binary Tree

- Representing tree nodes
  - Must contain both data and “pointers” to node’s children
  - Each node will be an object
- Array-based
  - Pointers will be array indices
- Link-based
  - Use C++ pointers

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# Array-Based Representation

- Class of array-based data members

```
TreeNode<ItemType>  tree[MAX_NODES];  // Array of tree nodes
int                 root;             // Index of root
int                 free;             // Index of free list
```

  - Variable root is index to tree's root node within the array tree
  - If tree is empty, root = -1

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# Array-Based Representation

- As tree changes (additions, removals) …
  - Nodes may not be in contiguous array elements
- Thus, need list of available nodes
  - Called a free list
- Insert node to tree
  - Obtain available node from free list
- Remove node from tree
  - Placed in free list for later use

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# Array-Based Representation

```
template<class ItemType>
class TreeNode
{
private:
   ItemType  item;        // Data portion
   int       leftChild;   // Index to left child
   int       rightChild;  // Index to right child

public:
   TreeNode();
   TreeNode(const ItemType& nodeItem, int left, int right);

// Declarations of the methods setItem, getItem, setLeft, getLeft,
// setRight, and getRight are here.

   . . .

}; // end TreeNode
```

LISTING 16-1 The class TreeNode for an array-based implementation of the ADT binary tree

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# Array-Based Representation

Is Figure 16-1 (a) a binary search tree?

- yes

Is Figure 16-1 (a) a balanced tree?

- yes

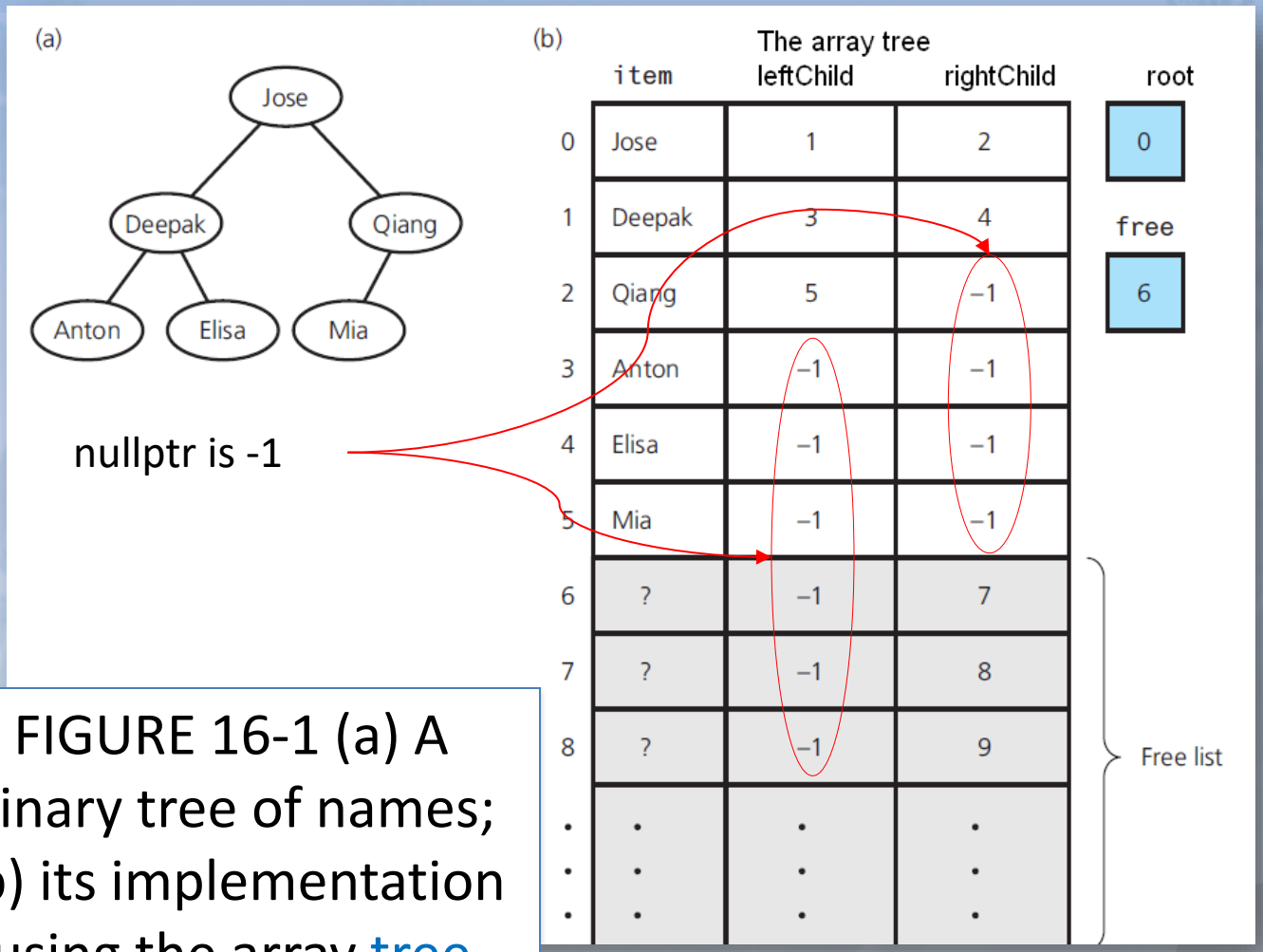


FIGURE 16-1 (a) A binary tree of names; (b) its implementation using the array tree

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# Link-Based Representation

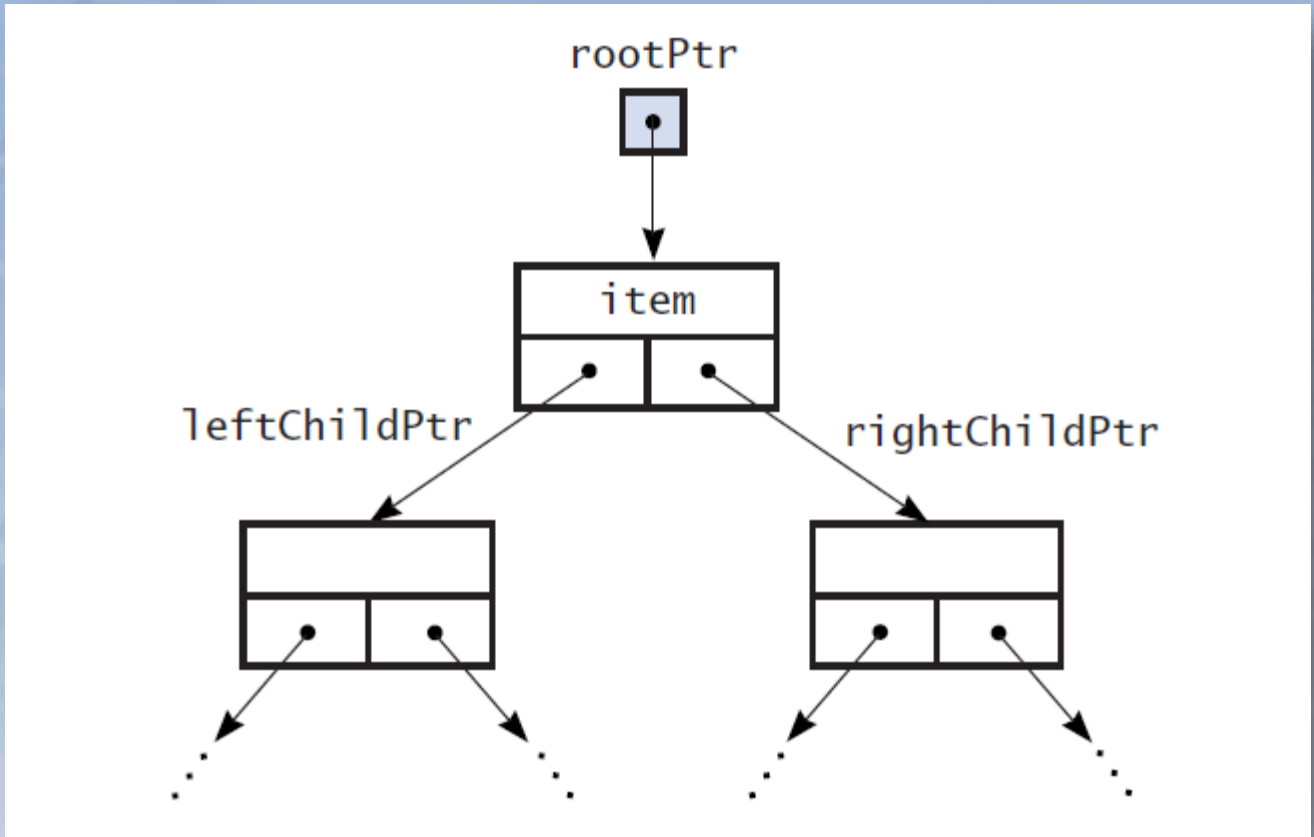


Figure 16-2 Most common way of implementing a tree

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# Link-Based Representation

```
/** A class of nodes for a link-based binary tree.
 @file BinaryNode.h */

#ifndef _BINARY_NODE
#define _BINARY_NODE

template<class ItemType>
class BinaryNode
{
private:
   ItemType              item;          // Data portion
   BinaryNode<ItemType>* leftChildPtr;  // Pointer to left child
   BinaryNode<ItemType>* rightChildPtr; // Pointer to right child

public:
   BinaryNode();
   BinaryNode(const ItemType& anItem);
   BinaryNode(const ItemType& anItem,
              BinaryNode<ItemType>* leftPtr,
              BinaryNode<ItemType>* rightPtr);

   void setItem(const ItemType& anItem);
   ItemType getItem() const;

   bool isLeaf() const;

   BinaryNode<ItemType>* getLeftChildPtr() const;
   BinaryNode<ItemType>* getRightChildPtr() const;

   void setLeftChildPtr(BinaryNode<ItemType>* leftPtr);
   void setRightChildPtr(BinaryNode<ItemType>* rightPtr);
}; // end BinaryNode

#include "BinaryNode.cpp"
#endif
```

- If the tree is empty, rootPtr contains nullptr, else it will point to the tree's root node.
- For a binary tree that is not empty, rootPtr->getLeftChildPtr() points to the root's left subtree, and rootPtr->getRightChildPtr() points to the root's right subtree.
- If either of these subtrees is empty, the pointer to it would be nullptr.

LISTING 16-2 The header file containing the class BinaryNode

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# The Header File

```
/** ADT binary tree: Link-based implementation.
 @file BinaryNodeTree.h */

#ifndef BINARY_NODE_TREE_
#define BINARY_NODE_TREE_

#include "BinaryTreeInterface.h"
#include "BinaryNode.h"
#include "PrecondViolatedExcept.h"
#include "NotFoundException.h"
#include <memory>

template<class ItemType>
class BinaryNodeTree : public BinaryTreeInterface<ItemType>
{
private:
   std::shared_ptr<BinaryNode<ItemType>> rootPtr;

protected:
//------------------------------------------------------------
//      Protected Utility Methods Section:
//      Recursive helper methods for the public methods.
//------------------------------------------------------------
   int getHeightHelper(std::shared_ptr<BinaryNode<ItemType>> subTreePtr) const;
   int getNumberOfNodesHelper(std::shared_ptr<BinaryNode<ItemType>> subTreePtr) const;

   // Recursively adds a new node to the tree in a left/right fashion to keep tree balanced.
   auto balancedAdd(std::shared_ptr<BinaryNode<ItemType>> subTreePtr,
                    std::shared_ptr<BinaryNode<ItemType>> newNodePtr);
```

The methods are protected (not private) so that derived classes can use them.

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Header File

```
    // Removes the target value from the tree.
    virtual auto removeValue(std::shared_ptr<BinaryNode<ItemType>> subTreePtr,
                                          const ItemType target, bool& isSuccessful);

    // Copies values up the tree to overwrite value in current node until
    // a leaf is reached; the leaf is then removed, since its value is stored in the parent.
    auto moveValuesUpTree(std::shared_ptr<BinaryNode<ItemType>> subTreePtr);

    // Recursively searches for target value.
    virtual auto findNode(std::shared_ptr<BinaryNode<ItemType>> treePtr,
                                        const ItemType& target, bool& isSuccessful) const;

    // Copies the tree rooted at treePtr and returns a pointer to the root of the copy.
    auto copyTree(const std::shared_ptr<BinaryNode<ItemType>> oldTreeRootPtr) const;

    // Recursively deletes all nodes from the tree.
    void destroyTree(std::shared_ptr<BinaryNode<ItemType>> subTreePtr);

    // Recursive traversal helper methods:
    void preorder(void visit(ItemType&), std::shared_ptr<BinaryNode<ItemType>> treePtr) const;
    void inorder(void visit(ItemType&), std::shared_ptr<BinaryNode<ItemType>> treePtr) const;
    void postorder(void visit(ItemType&), std::shared_ptr<BinaryNode<ItemType>> treePtr) const;

public:
//------------------------------------------------------------
//       Constructor and Destructor Section.
//------------------------------------------------------------
    BinaryNodeTree();
    BinaryNodeTree(const ItemType& rootItem);
    BinaryNodeTree(const ItemType& rootItem,
                   const std::shared_ptr<BinaryNodeTree<ItemType>> leftTreePtr,
                   const std::shared_ptr<BinaryNodeTree<ItemType>> rightTreePtr);
    BinaryNodeTree(const std::shared_ptr<BinaryNodeTree<ItemType>>& tree);
    virtual ~BinaryNodeTree();
```

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Header File

```
//-----------------------------------------------------------
//       Public BinaryTreeInterface Methods Section.
//-----------------------------------------------------------
   bool isEmpty() const;
   int getHeight() const;
   int getNumberOfNodes() const;
   ItemType getRootData() const throw(PrecondViolatedExcept);
   void setRootData(const ItemType& newData);
   bool add(const ItemType& newData); // Adds an item to the tree
   bool remove(const ItemType& data); // Removes specified item from the tree
   void clear();
   ItemType getEntry(const ItemType& anEntry) const throw(NotFoundException);
   bool contains(const ItemType& anEntry) const;

//-----------------------------------------------------------
//       Public Traversals Section.
//-----------------------------------------------------------
   void preorderTraverse(void visit(ItemType&)) const;
   void inorderTraverse(void visit(ItemType&)) const;
   void postorderTraverse(void visit(ItemType&)) const;

//-----------------------------------------------------------
//       Overloaded Operator Section.
//-----------------------------------------------------------
   BinaryNodeTree& operator=(const BinaryNodeTree& rightHandSide);
}; // end BinaryNodeTree

#include "BinaryNodeTree.cpp"
#endif
```

Public traversal methods each call a protected method that performs the actual recursion

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Implementation

**The constructors.** The public constructors have the following definitions in the implementation file:

```
template<class ItemType>
BinaryNodeTree<ItemType>::BinaryNodeTree() : rootPtr(nullptr)
{
} // end default constructor

template<class ItemType>
BinaryNodeTree<ItemType>::BinaryNodeTree(const ItemType& rootItem)
{
   rootPtr = new BinaryNode<ItemType>(rootItem, nullptr, nullptr);
}  // end constructor

template<class ItemType>
BinaryNodeTree<ItemType>::BinaryNodeTree(const ItemType& rootItem,
                                         const BinaryNodeTree<ItemType>* leftTreePtr,
                                         const BinaryNodeTree<ItemType>* rightTreePtr)
{
   rootPtr = new BinaryNode<ItemType>(rootItem,
                                      copyTree(leftTreePtr->rootPtr),
                                      copyTree(rightTreePtr->rootPtr));
}  // end constructor
```

set the value of rootPtr to nullptr

sets rootPtr to point to a new node containing the given data item

sets rootPtr to point to a new node containing the given data item and pointers to copies of the given subtrees

Constructors

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# The Implementation

```
template<class ItemType>
std::shared_ptr<BinaryNode<ItemType>> BinaryNodeTree<ItemType>::copyTree(
     const std::shared_ptr<BinaryNode<ItemType>> oldTreeRootPtr) const
{
   std::shared_ptr<BinaryNode<ItemType>> newTreePtr;

   // Copy tree nodes during a preorder traversal
   if (oldTreeRootPtr != nullptr)
   {
      // Copy node
      newTreePtr = std::make_shared<BinaryNode<ItemType>>(oldTreeRootPtr->getItem(),
                                                          nullptr, nullptr);
      newTreePtr->setLeftChildPtr(copyTree(oldTreeRootPtr->getLeftChildPtr()));
      newTreePtr->setRightChildPtr(copyTree(oldTreeRootPtr->getRightChildPtr()));
   }  // end if
   // Else tree is empty (newTreePtr is nullptr)

   return newTreePtr;
}  // end copyTree
```

To copy a tree, traverse it in **preorder** and insert each item visited into a new node

Protected method copyTree called by copy constructor

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# The Implementation

To delete a tree, traverse it in **postorder** and delete

```
template<class ItemType>
void BinaryNodeTree<ItemType>::
     destroyTree(std::shared_ptr<BinaryNode<ItemType>> subTreePtr)
{
   if (subTreePtr != nullptr)
   {
      destroyTree(subTreePtr->getLeftChildPtr());
      destroyTree(subTreePtr->getRightChildPtr());
      subTreePtr.reset(); // Decrement reference count to node
   }  // end if
}  // end destroyTree
```

destroyTree used by destructor which simply calls this method

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# The Implementation

```
template<class ItemType>
int BinaryNodeTree<ItemType>::
    getHeightHelper(std::shared_ptr<BinaryNode<ItemType>> subTreePtr) const
{
   if (subTreePtr == nullptr)
      return 0;
   else
      return 1 + max(getHeightHelper(subTreePtr->getLeftChildPtr()),
                     getHeightHelper(subTreePtr->getRightChildPtr()));
}  // end getHeightHelper
```

Protected method getHeightHelper

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# The Implementation

```
template<class ItemType>
void BinaryNodeTree<ItemType>::
     inorder(void visit(ItemType&),
             std::shared_ptr<BinaryNode<ItemType>> treePtr) const
{
   if (treePtr != nullptr)
   {
      inorder(visit, treePtr->getLeftChildPtr());
      ItemType theItem = treePtr->getItem();
      visit(theItem);
      inorder(visit, treePtr->getRightChildPtr());
   }  // end if
}  // end inorder
```

Protected method that enables recursive traversals.

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# Link-Based Implementation of the ADT Binary Search Tree

- Uses same node objects as for binary-tree implementation.
- Class BinaryNode from Listing16-2 will be used
- Recursive search algorithm from Section15.3.2 is basis for operations

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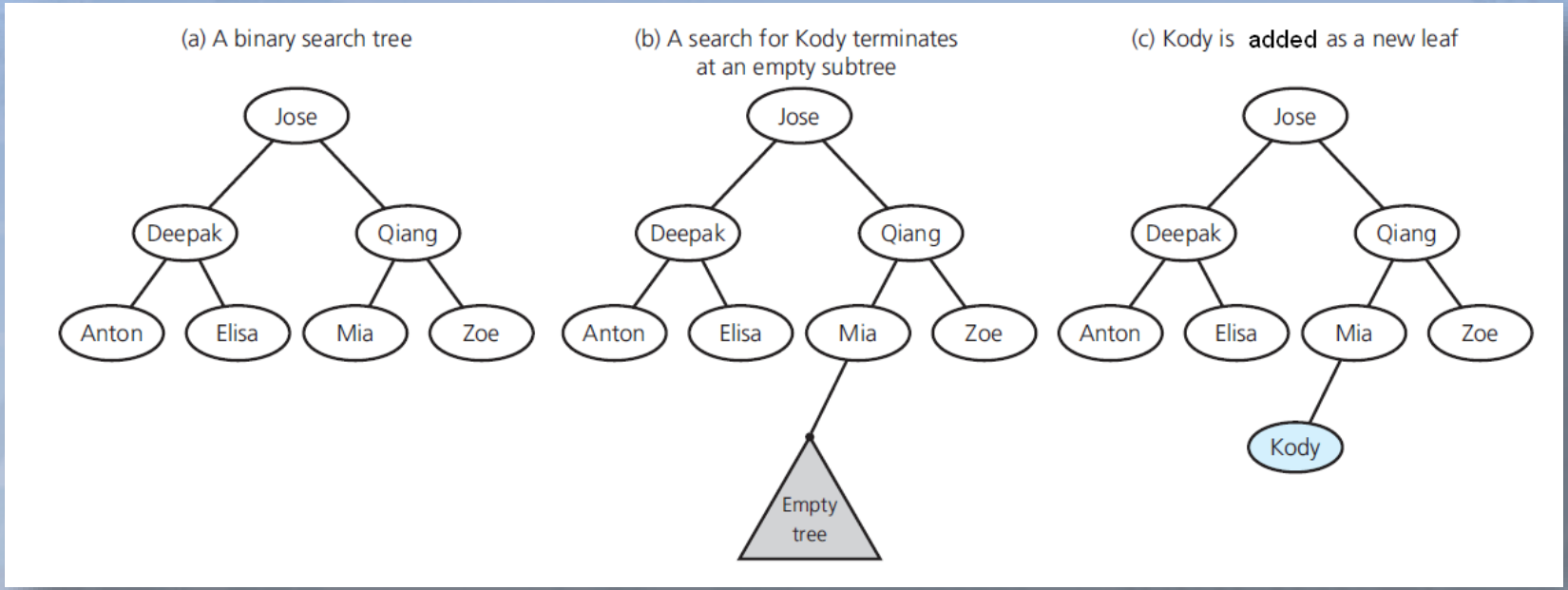
# Link-Based Implementation of the ADT Binary Search Tree




- If you take the output of a preorder traversal of a binary search tree and use it with insertInorder to create a binary search tree, you will obtain a duplicate tree. Tree (a) was used in previous chapter.
- The copy constructor for the ADT binary tree used a preorder traversal to copy the tree.
- To insert Kody, search the BST, and search algorithm would terminate at Mia's left child

FIGURE 16-7 Adding Kody to a binary search tree

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# Link-Based Implementation of the ADT Binary Search Tree

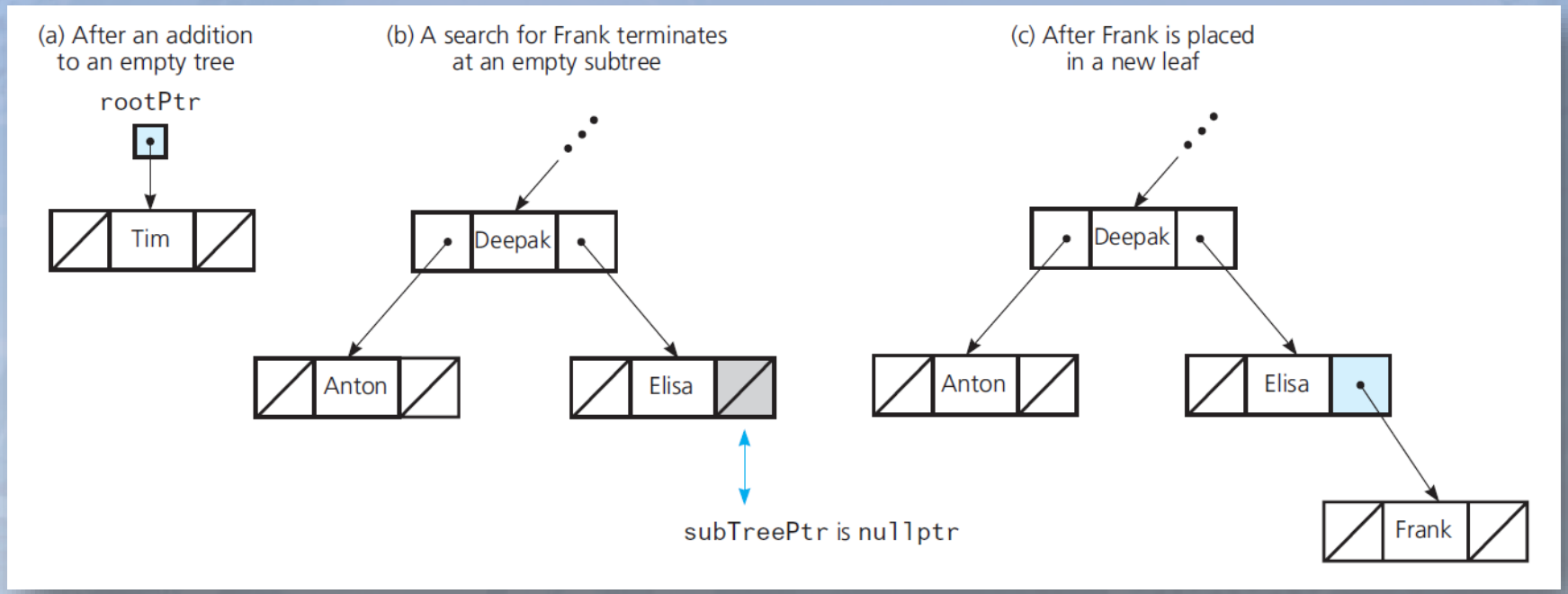


Figure 16-8 Adding new data to a binary search tree

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# Link-Based Implementation of the ADT Binary Search Tree

3 cases for removing node *N*:

1. *N* is a leaf
   - Remove leaf containing target
   - Set pointer in parent to nullptr
2. *N* has only left (or right) child – cases are symmetrical
   - After *N* removed, all data items rooted at *L* (or *R*) are adopted by root of *N*
   - All items adopted are in correct order, binary search tree property preserved
3. *N* has two children
   - Locate another node *M* easier to remove from tree than *N*
   - Copy item that is in *M* to *N*
   - Remove *M* from tree

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# Link-Based Implementation of the ADT Binary Search Tree

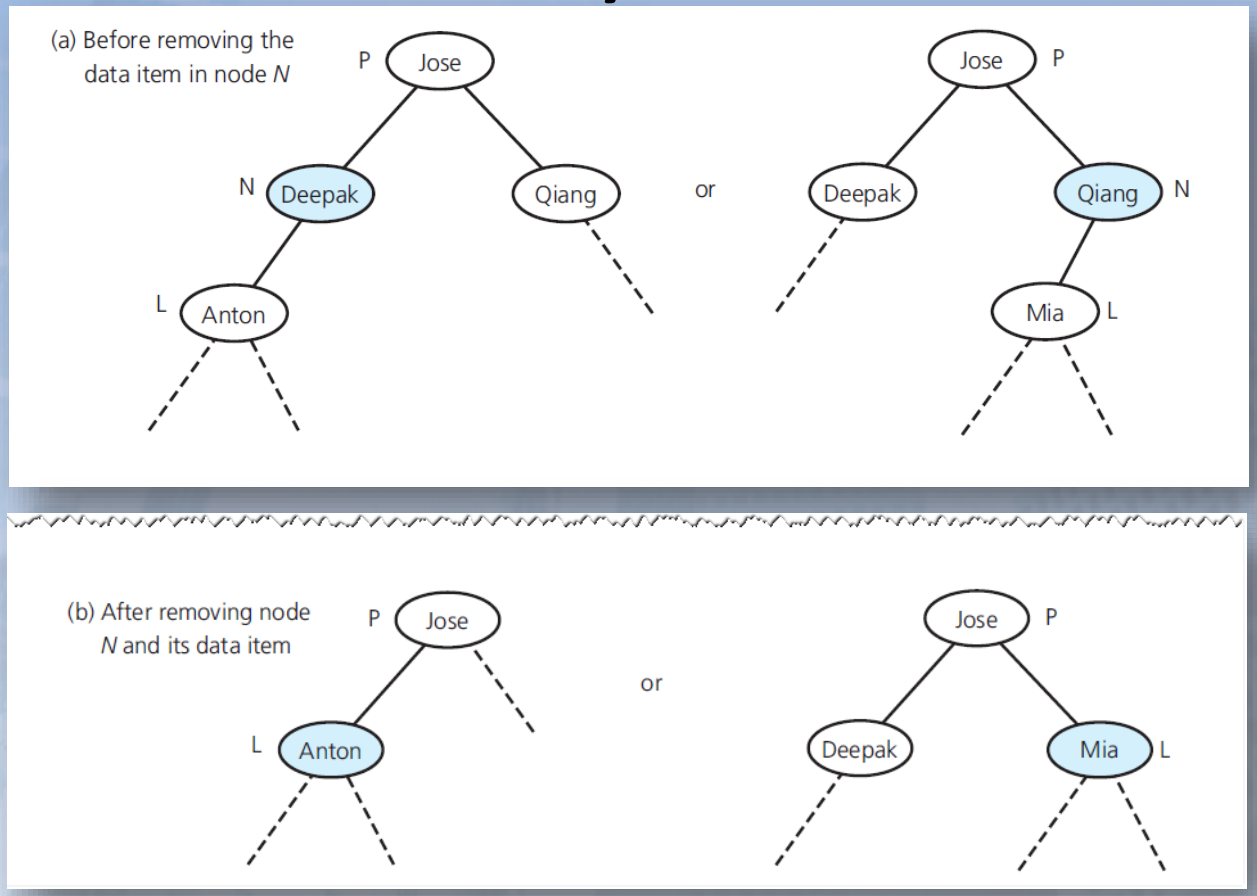


Figure 16-9 Case 2 for removeValue: The data item to remove is in a node N that has only a left child and whose parent is node P

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# Link-Based Implementation of the ADT Binary Search Tree

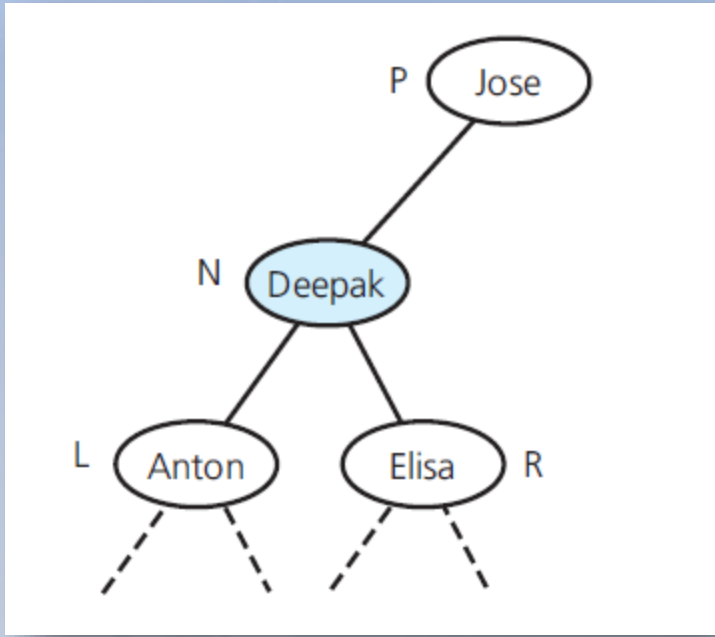


FIGURE 16-10 Case 3: The data item to remove is in a node *N* that has two children

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# Link-Based Implementation of the ADT Binary Search Tree

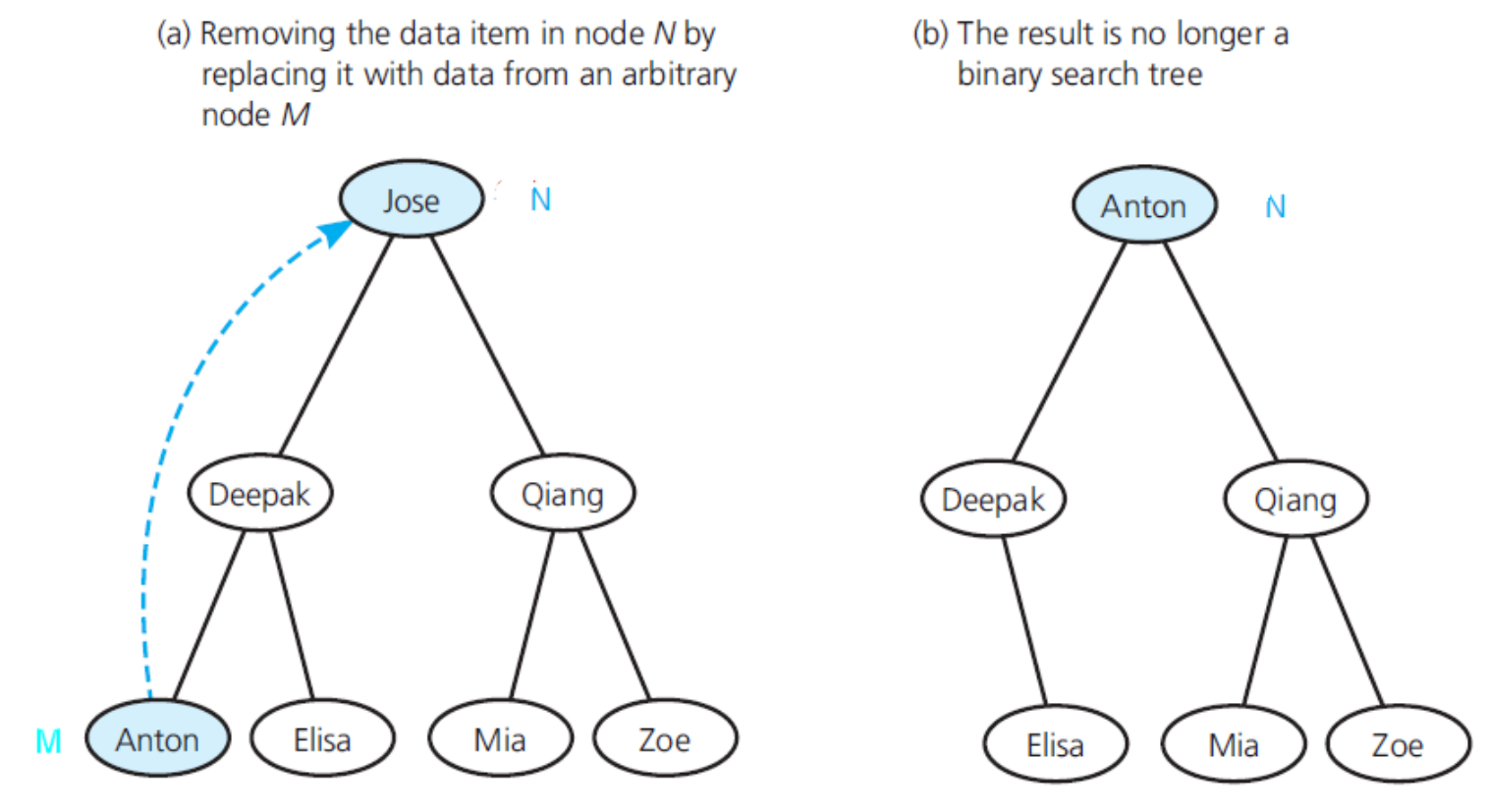


FIGURE16-11 Not any node will do

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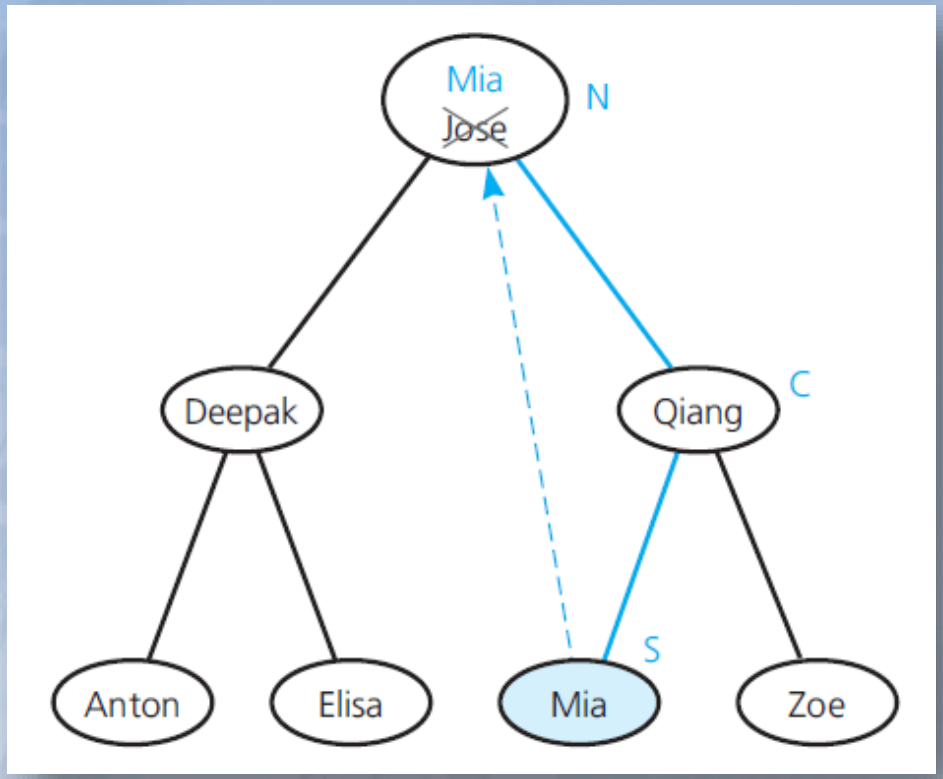
# Link-Based Implementation of the ADT Binary Search Tree




- inorder successor of a data item is the leftmost node of N's right subtree.

FIGURE 16-13 Replacing the data item in node *N* with its **inorder successor**

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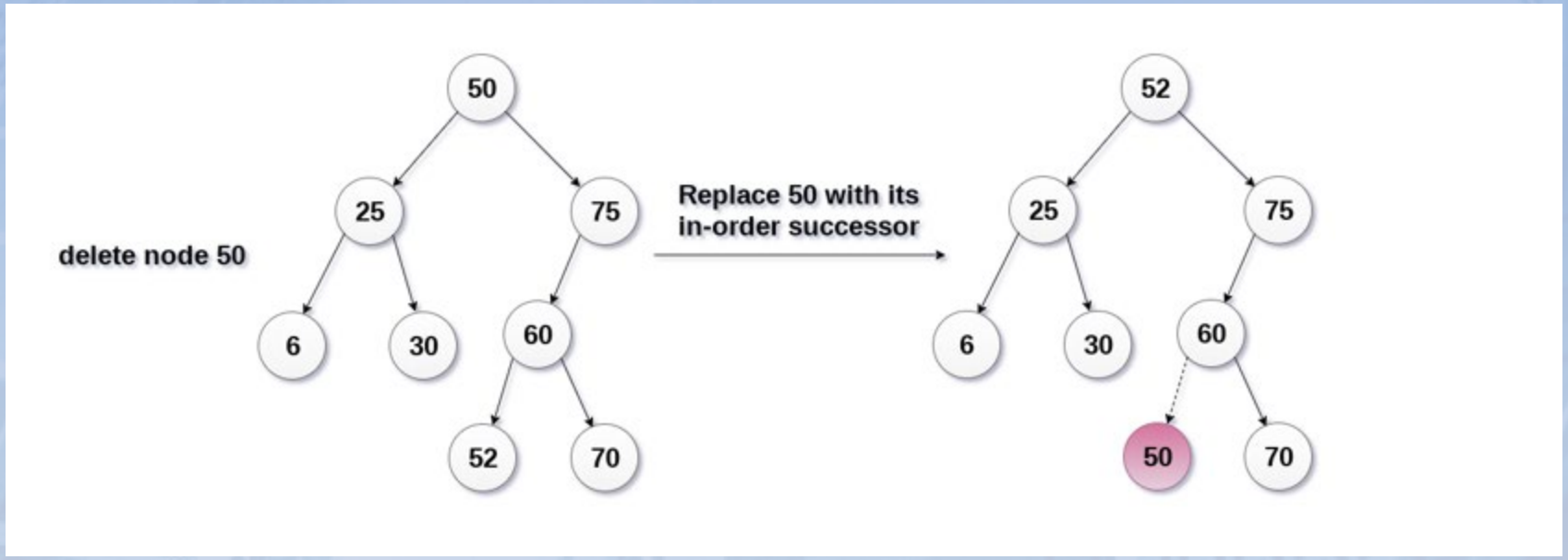
# Link-Based Implementation of the ADT Binary Search Tree




https://www.javatpoint.com/deletion-in-binary-search-tree

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# Saving a Binary Search Tree in a File

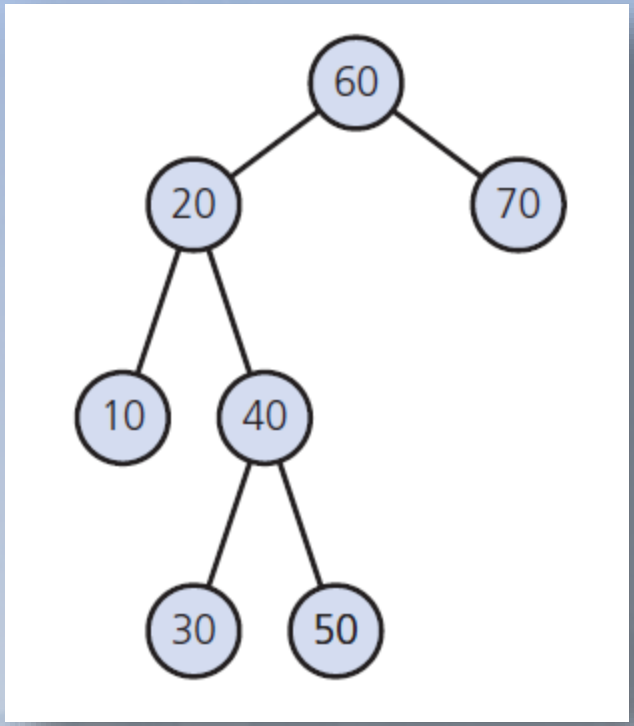


FIGURE 16-15 An initially empty binary search tree after the addition of 60, 20, 10, 40, 30, 50, and 70

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# Saving a Binary Search Tree in a File

- Use preorder traversal to save binary search tree in a file
  - Restore to original shape by using method add
- Balanced binary search tree increases efficiency of ADT operations

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# Saving a Binary Search Tree in a File

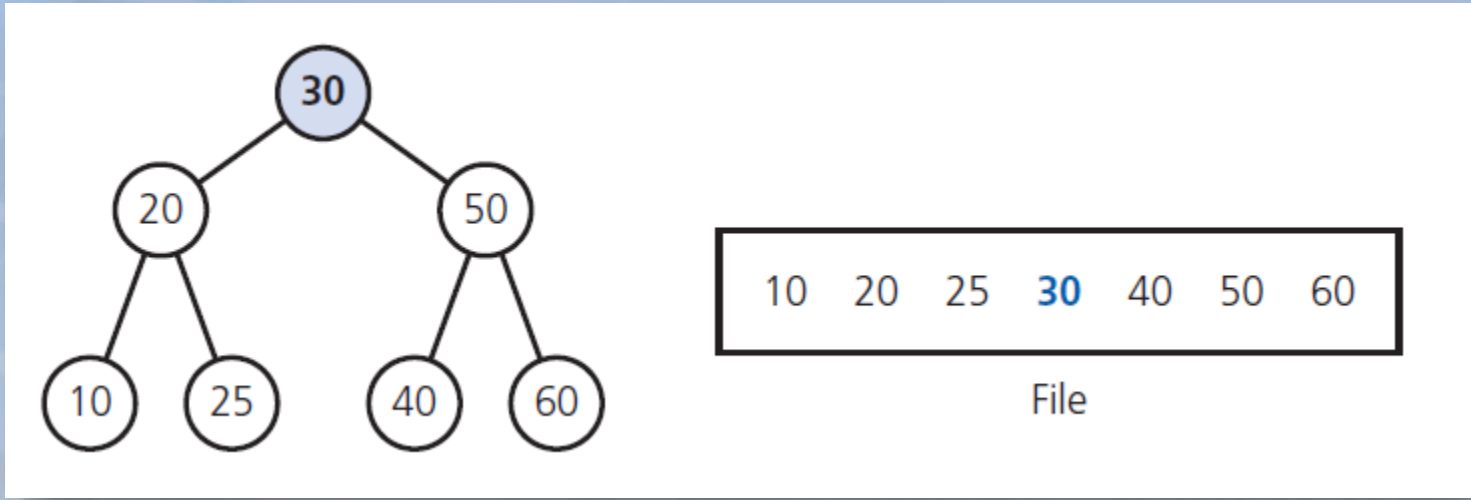


FIGURE 16-16 A full tree saved in a file by using inorder traversal

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# Saving a Binary Search Tree in a File

```
// Builds a minimum-height binary search tree from n sorted values in a file.
// Returns a pointer to the tree's root.
readTree(n: integer): BinaryNodePointer

   if (n > 0)
   {
      // Get the root
      treePtr = pointer to new node with nullptr as its child pointers
      rootItem = next item from file
      treePtr->setItem(rootItem)

      // Construct the left subtree
      leftPtr = readFullTree(treePtr->getLeftChildPtr(), n / 2)
      treePtr->setLeftChildPtr(leftPtr)

      // Construct the right subtree
      rightPtr = readFullTree(treePtr->getRightChildPtr(), (n - 1) / 2)
      treePtr->setRightChildPtr(rightPtr )

      return treePtr
   }
   else
      return nullptr
```

Building a minimum-height binary search tree by reading sorted data sequentially from file

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# Tree Sort

```
// Sorts the integers in an array into ascending order.
treeSort(anArray: array, n: integer)
{
    Add anArray's entries to a binary search tree bst
    Traverse bst in inorder. As you visit bst's nodes, copy their data items into successive
      locations of anArray
}
```

- Tree sort:
  - Insertion is O(log n) in the average case, and O(n) in the worse case
  - Thus, insertion is O(n × log n) in the average case, and $O(n^2)$ in the worst case
  - Traversal involves one copy: O(n)
  - Thus, traversal is O(n × log n) in the average case, and $O(n^2)$ in the worst case

Tree sort uses a binary search tree.

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# General Trees

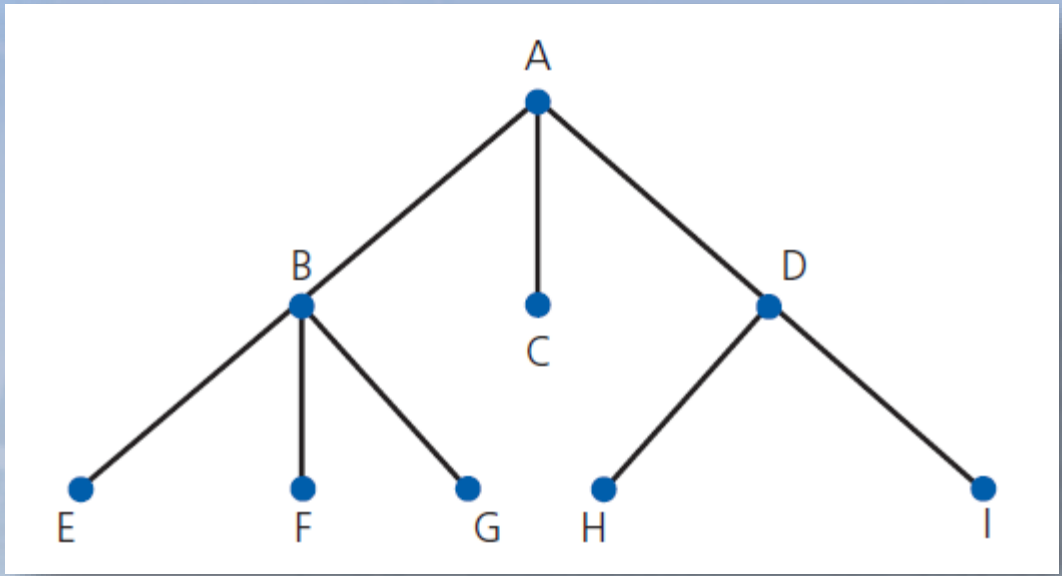


FIGURE 16-18 A general tree or an *n*-ary tree with n = 3

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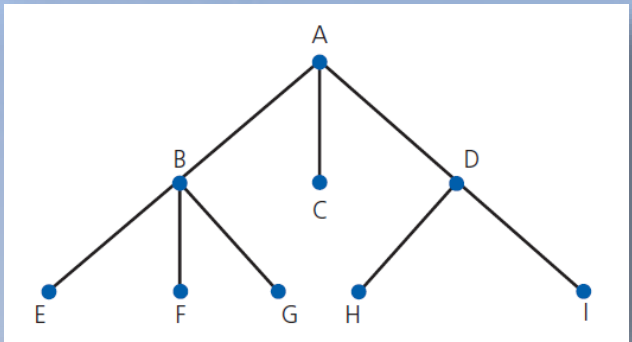
# General Trees



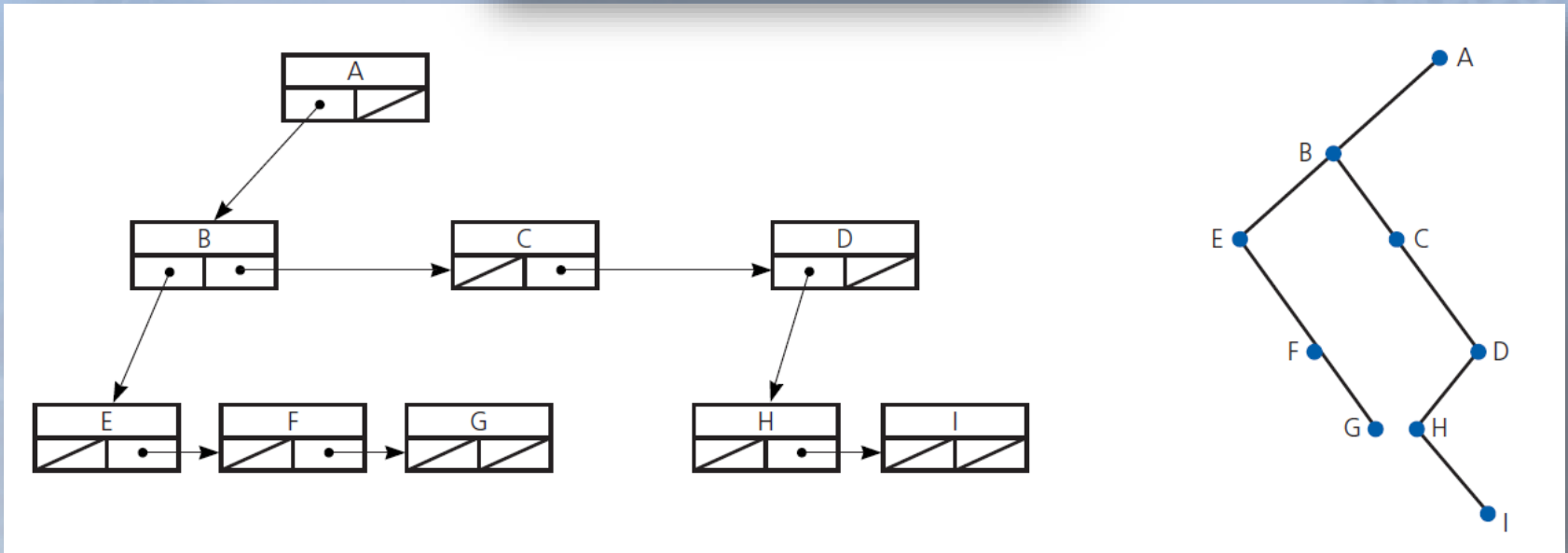




FIGURE 16-19 An implementation of a general tree and its equivalent binary tree

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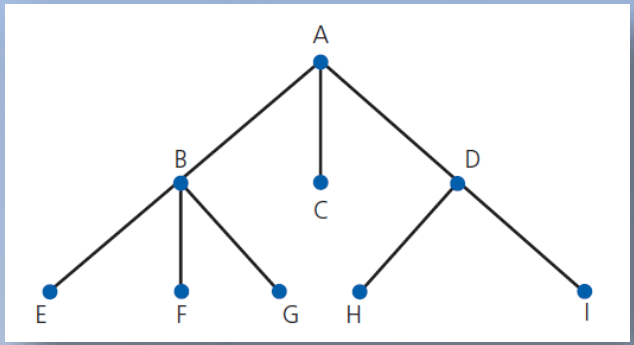
# General Trees



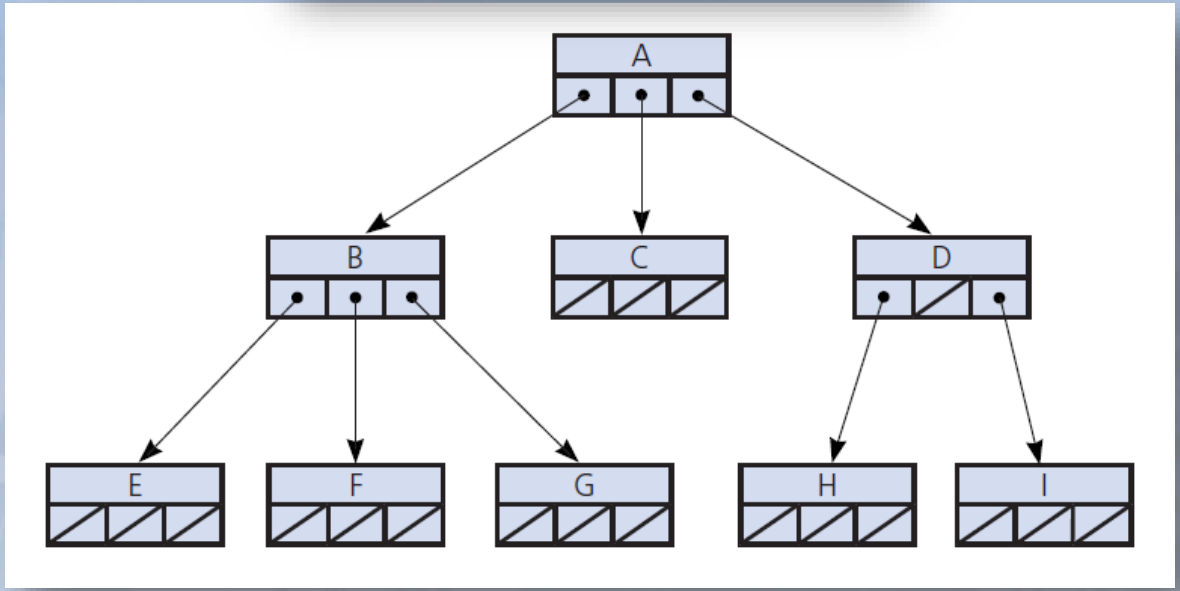




FIGURE 16-20 An implementation of the *n*-ary tree in Figure 16-18

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# End

Chapter 16

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# Appendix

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# Link-Based Implementation of the ADT Binary Search Tree

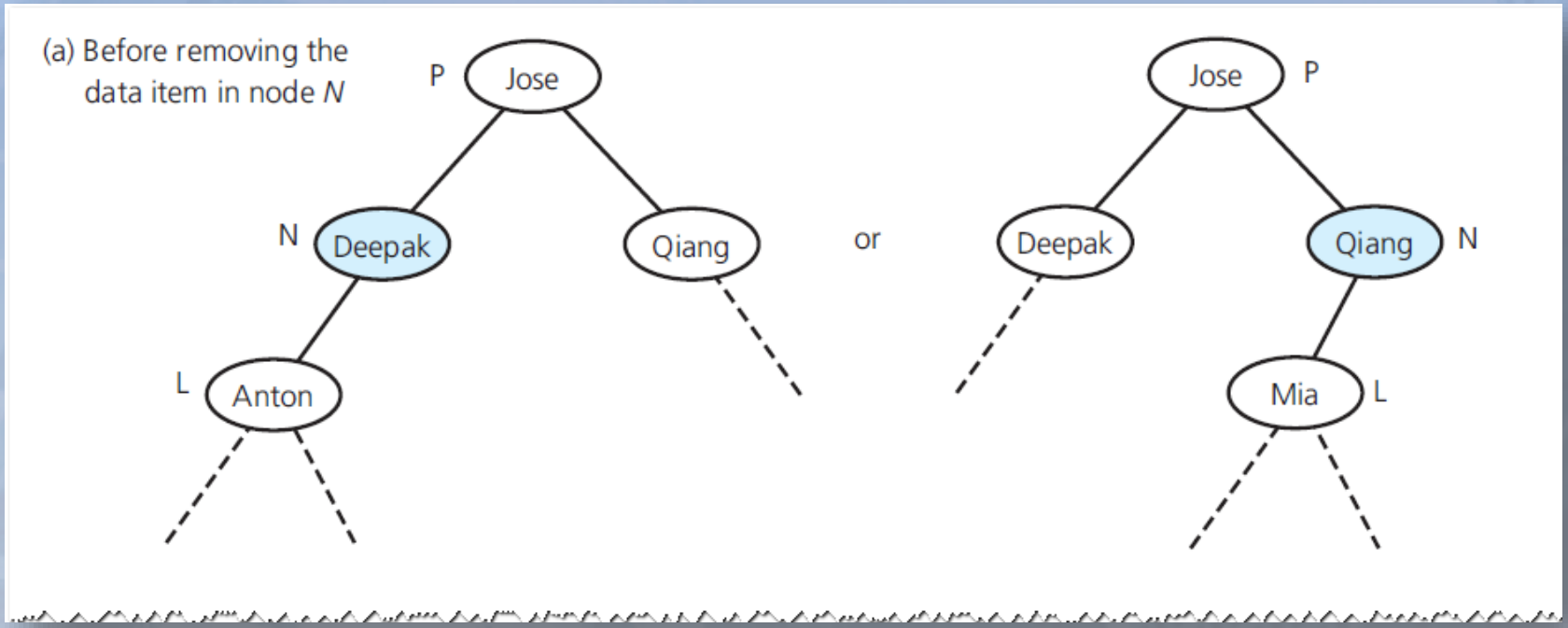


FIGURE 16-9 Case 2 for removeValue: The data item to remove is in a node *N* that has only a left child and whose parent is node *P*

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# Link-Based Implementation of the ADT Binary Search Tree

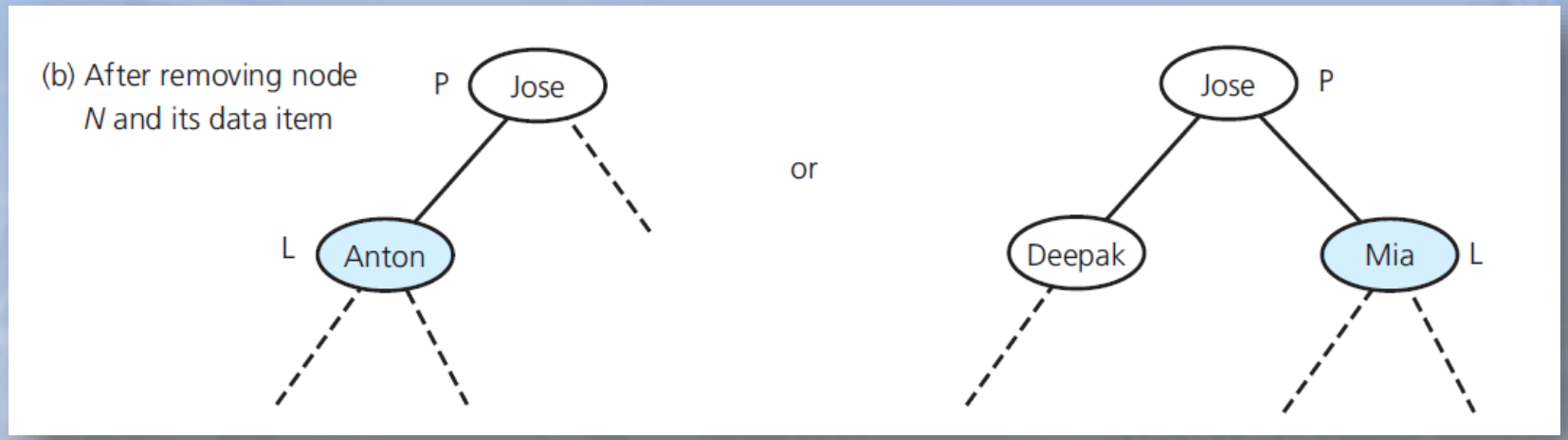


Figure 16-9 Case 2 for removeValue: The data item to remove is in a node N that has only a left child and whose parent is node P

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# Link-Based Implementation of the ADT Binary Search Tree

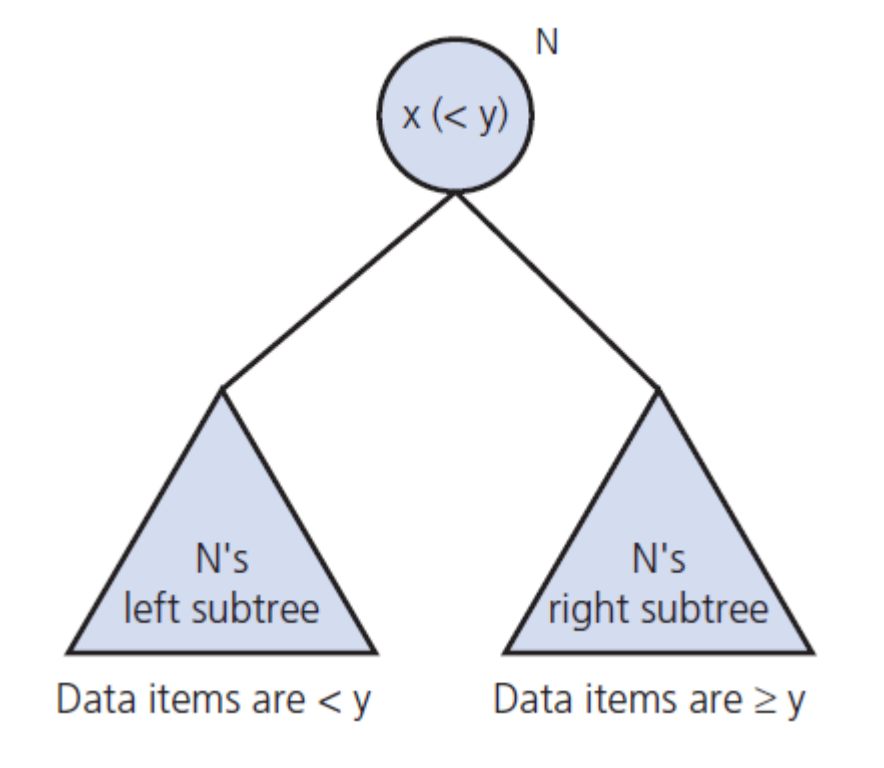


FIGURE 16-12

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# Link-Based Representation

```
/** A class of nodes for a link-based binary tree.
 @file BinaryNode.h */

#ifndef _BINARY_NODE
#define _BINARY_NODE

template<class ItemType>
class BinaryNode
{
private:
   ItemType              item;          // Data portion
   BinaryNode<ItemType>* leftChildPtr;  // Pointer to left child
   BinaryNode<ItemType>* rightChildPtr; // Pointer to right child

public:
   BinaryNode();
   BinaryNode(const ItemType& anItem);
   BinaryNode(const ItemType& anItem,
              BinaryNode<ItemType>* leftPtr,
              BinaryNode<ItemType>* rightPtr);
```

- If the tree is empty, rootPtr contains nullptr .
- For a binary tree that is not empty, rootPtr->getLeftChildPtr() points to the root's left subtree, and rootPtr->getRightChildPtr() points to the root's right subtree.
- If either of these subtrees is empty, the pointer to it would be nullptr

LISTING 16-2 The header file containing the class BinaryNode for a link-based implementation of the ADT binary tree

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# Link-Based Representation

```
   void setItem(const ItemType& anItem);
   ItemType getItem() const;

   bool isLeaf() const;

   BinaryNode<ItemType>* getLeftChildPtr() const;
   BinaryNode<ItemType>* getRightChildPtr() const;

   void setLeftChildPtr(BinaryNode<ItemType>* leftPtr);
   void setRightChildPtr(BinaryNode<ItemType>* rightPtr);
}; // end BinaryNode

#include "BinaryNode.cpp"
#endif
```

LISTING 16-2 (cont.) The header file containing the class BinaryNode for a link-based implementation of the ADT binary tree

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# The Header File

```
protected:
//-----------------------------------------------------------
//      Protected Utility Methods Section:
//      Recursive helper methods for the public methods.
//-----------------------------------------------------------
   int getHeightHelper(std::shared_ptr<BinaryNode<ItemType>> subTreePtr) const;
   int getNumberOfNodesHelper(std::shared_ptr<BinaryNode<ItemType>> subTreePtr) const;

   // Recursively adds a new node to the tree in a left/right fashion to keep tree balanced.
   auto balancedAdd(std::shared_ptr<BinaryNode<ItemType>> subTreePtr,
                    std::shared_ptr<BinaryNode<ItemType>> newNodePtr);
```

The methods are protected (not private) so that derived classes can use them.

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Header File

```
    // Removes the target value from the tree.
    virtual auto removeValue(std::shared_ptr<BinaryNode<ItemType>> subTreePtr,
                                          const ItemType target, bool& isSuccessful);

    // Copies values up the tree to overwrite value in current node until
    // a leaf is reached; the leaf is then removed, since its value is stored in the parent.
    auto moveValuesUpTree(std::shared_ptr<BinaryNode<ItemType>> subTreePtr);

    // Recursively searches for target value.
    virtual auto findNode(std::shared_ptr<BinaryNode<ItemType>> treePtr,
                                        const ItemType& target, bool& isSuccessful) const;

    // Copies the tree rooted at treePtr and returns a pointer to the root of the copy.
    auto copyTree(const std::shared_ptr<BinaryNode<ItemType>> oldTreeRootPtr) const;

    // Recursively deletes all nodes from the tree.
    void destroyTree(std::shared_ptr<BinaryNode<ItemType>> subTreePtr);
```

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Header File

```
    // Recursive traversal helper methods:
    void preorder(void visit(ItemType&), std::shared_ptr<BinaryNode<ItemType>> treePtr) const;
    void inorder(void visit(ItemType&), std::shared_ptr<BinaryNode<ItemType>> treePtr) const;
    void postorder(void visit(ItemType&), std::shared_ptr<BinaryNode<ItemType>> treePtr) const;

public:
//------------------------------------------------------------
//       Constructor and Destructor Section.
//------------------------------------------------------------
    BinaryNodeTree();
    BinaryNodeTree(const ItemType& rootItem);
    BinaryNodeTree(const ItemType& rootItem,
                   const std::shared_ptr<BinaryNodeTree<ItemType>> leftTreePtr,
                   const std::shared_ptr<BinaryNodeTree<ItemType>> rightTreePtr);
    BinaryNodeTree(const std::shared_ptr<BinaryNodeTree<ItemType>>& tree);
    virtual ~BinaryNodeTree();
```

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Header File

```
//-----------------------------------------------------------
//       Public BinaryTreeInterface Methods Section.
//-----------------------------------------------------------
   bool isEmpty() const;
   int getHeight() const;
   int getNumberOfNodes() const;
   ItemType getRootData() const throw(PrecondViolatedExcept);
   void setRootData(const ItemType& newData);
   bool add(const ItemType& newData); // Adds an item to the tree
   bool remove(const ItemType& data); // Removes specified item from the tree
   void clear();
   ItemType getEntry(const ItemType& anEntry) const throw(NotFoundException);
   bool contains(const ItemType& anEntry) const;
```

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Header File

```
//-----------------------------------------------------------
//       Public Traversals Section.
//-----------------------------------------------------------
   void preorderTraverse(void visit(ItemType&)) const;
   void inorderTraverse(void visit(ItemType&)) const;
   void postorderTraverse(void visit(ItemType&)) const;

//-----------------------------------------------------------
//       Overloaded Operator Section.
//-----------------------------------------------------------
   BinaryNodeTree& operator=(const BinaryNodeTree& rightHandSide);
}; // end BinaryNodeTree

#include "BinaryNodeTree.cpp"
#endif
```

Public traversal methods each call a protected method that performs the actual recursion

LISTING 16-3 A header file for the link-based implementation of the class BinaryNodeTree

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# The Implementation

```
template<class ItemType>
bool BinaryNodeTree<ItemType>::add(const ItemType& newData)
{
   auto newNodePtr = std::make_shared<BinaryNode<ItemType>>(newData);
   rootPtr = balancedAdd(rootPtr, newNodePtr);

   return true;
}  // end add
```

Method add

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# The Implementation

```
template<class ItemType>
BinaryNode<ItemType>* BinaryNodeTree<ItemType>::
                      balancedAdd(BinaryNode<ItemType>* subTreePtr,
                                  BinaryNode<ItemType>* newNodePtr)
{
   if (subTreePtr == nullptr)
      return newNodePtr;
   else
   {
      BinaryNode<ItemType>* leftPtr = subTreePtr->getLeftChildPtr();
      BinaryNode<ItemType>* rightPtr = subTreePtr->getRightChildPtr();

      if (getHeightHelper(leftPtr) > getHeightHelper(rightPtr))
      {
         rightPtr = balancedAdd(rightPtr, newNodePtr);
         subTreePtr->setRightChildPtr(rightPtr);
      }
      else
      {
         leftPtr = balancedAdd(leftPtr, newNodePtr);
         subTreePtr->setLeftChildPtr(leftPtr);
      }  // end if

      return subTreePtr;
   }  // end if
}  // end balancedAdd
```

Method balancedAdd

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# The Implementation

```
// Nonrecursively traverses a binary tree in inorder.
traverse(visit(item: ItemType): void): void
{
   // Initialize
   nodeStack = A new, empty stack
   curPtr = rootPtr // Start at root
   done = false

   while (!done)
   {
      if (curPtr != nullptr)
      {
         // Place pointer to node on stack before traversing the node's left subtree
         nodeStack.push(curPtr)

         // Traverse the left subtree
         curPtr = curPtr->getLeftChildPtr()
      }
      else // Backtrack from the empty subtree and visit the node at the top of
```

Nonrecursive inorder traversal

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# The Implementation

```
    }
    else   // Backtrack from the empty subtree and visit the node at the top of
           // the stack; however, if the stack is empty, you are done
    {
       done = nodeStack.isEmpty()
       if (!done)
       {
          nodeStack.peek(curPtr)
          visit(curPtr->getItem())
          nodeStack.pop()

          // Traverse the right subtree of the node just visited
          curPtr = curPtr ->getRightChildPtr()
       }
    }
  }
}
```

Nonrecursive inorder traversal

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# Link-Based Implementation of the ADT Binary Search Tree

```
// Removes the given target from the binary search tree to which subTreePtr points.
// Returns a pointer to the node at this tree location after the value is removed.
// Sets isSuccessful to true if the removal is successful, or false otherwise.
removeValue(subTreePtr: BinaryNodePointer, target: ItemType,
            isSuccessful: boolean&): BinaryNodePointer
{
   if (subTreePtr == nullptr)
   {
      isSuccessful = false
   }
   else if (subTreePtr->getItem() == target)
   {
      // Item is in the root of some subtree
      subTreePtr = removeNode(subTreePtr) // Remove the item
      isSuccessful = true
   }
   else if (subTreePtr->getItem() > target)
   {
      // Search the left subtree
      tempPtr = removeValue(subTreePtr->getLeftChildPtr(), target, isSuccessful)
      subTreePtr->setLeftChildPtr(tempPtr)
   }
   else
```

Final draft of the removal algorithm

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# Link-Based Implementation of the ADT Binary Search Tree

```
    }
    else
    {
        // Search the right subtree
        tempPtr = removeValue(subTreePtr->getRightChildPtr(), target, isSuccessful)
        subTreePtr->setRightChildPtr(tempPtr)
    }
    return subTreePtr
}

// Removes the data item in the node, N, to which nodePtr points.
// Returns a pointer to the node at this tree location after the removal.
removeNode(nodePtr: BinaryNodePointer): BinaryNodePointer
{
    if (N is a leaf)
    {
        // Remove leaf from the tree
        Delete the node to which nodePtr points (done for us if nodePtr is a smart pointer)
        return nodePtr
    }
    else if (N has only one child C)
```

Final draft of the removal algorithm

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# Link-Based Implementation of the ADT Binary Search Tree

```
}
else if (N has only one child C)
{
   // C replaces N as the child of N's parent
   if (C is a left child)
      nodeToConnectPtr = nodePtr->getLeftChildPtr()
   else
      nodeToConnectPtr = nodePtr->getRightChildPtr()

   Delete the node to which nodePtr points (done for us if nodePtr is a smart pointer)
   return nodeToConnectPtr
}
```

Final draft of the removal algorithm

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# Link-Based Implementation of the ADT Binary Search Tree

```
    else // N has two children
    {
        // Find the inorder successor of the entry in N: it is in the left subtree rooted
        // at N's right child
        tempPtr = removeLeftmostNode(nodePtr->getRightChildPtr(), newNodeValue)
        nodePtr->setRightChildPtr(tempPtr)
        nodePtr->setItem(newNodeValue) // Put replacement value in node N
        return nodePtr
    }
}

// Removes the leftmost node in the left subtree of the node pointed to by nodePtr.
// Sets inorderSuccessor to the value in this node.
// Returns a pointer to the revised subtree.
removeLeftmostNode(nodePtr: BinaryNodePointer,
                   inorderSuccessor: ItemType&): BinaryNodePointer
{
```

Final draft of the removal algorithm

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# Link-Based Implementation of the ADT Binary Search Tree

```
                    inorderSuccessor: ItemType&): BinaryNodePointer
{
   if (nodePtr->getLeftChildPtr() == nullptr)
   {
      // This is the node you want; it has no left child, but it might have a right subtree
      inorderSuccessor = nodePtr->getItem()
      return removeNode(nodePtr)
   }
   else
   {
      tempPtr = removeLeftmostNode(nodePtr->getLeftChildPtr(), inorderSuccessor)
      nodePtr->setLeftChildPtr(tempPtr)
      return nodePtr
   }
}
```

Final draft of the removal algorithm

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# The Implementation

```
template<class ItemType>
BinaryNodeTree<ItemType>::
                    BinaryNodeTree(const BinaryNodeTree<ItemType>& treePtr)
{
   rootPtr = copyTree(treePtr.rootPtr);
}  // end copy constructor
```

Copy constructor

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# Link-Based Implementation of the ADT Binary Search Tree

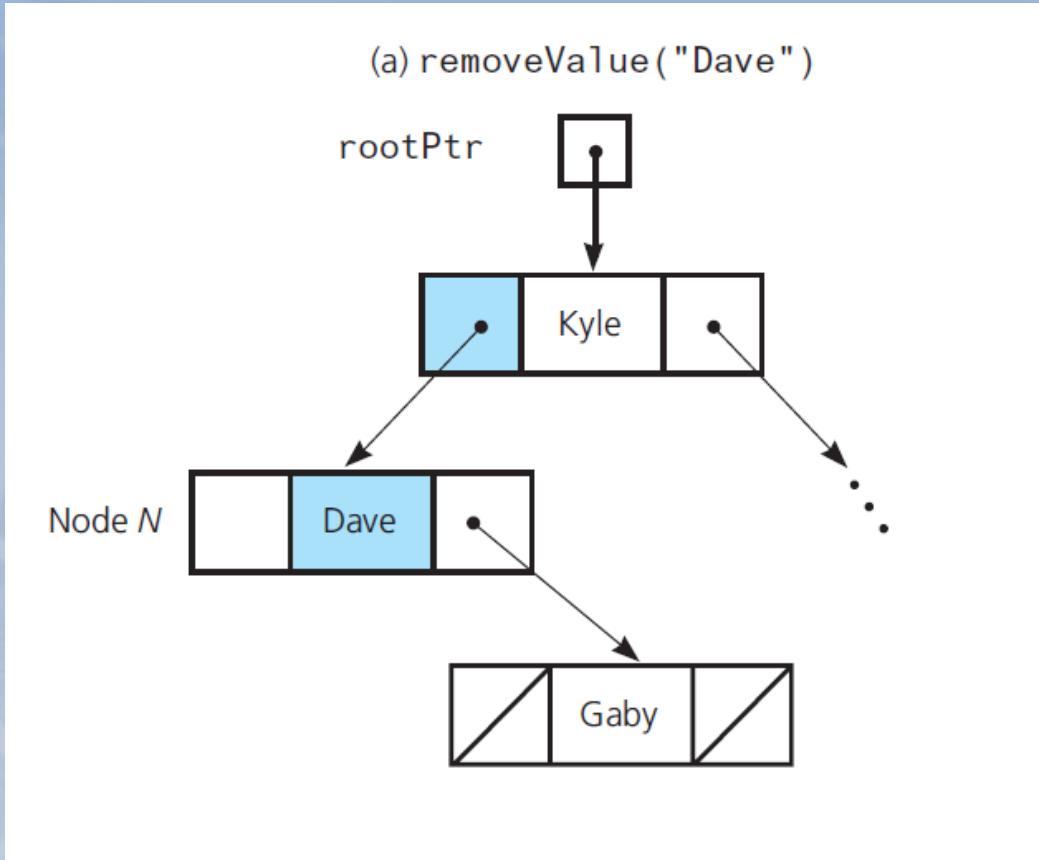


FIGURE 16-14 Recursive removal of node N

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# Link-Based Implementation of the ADT Binary Search Tree

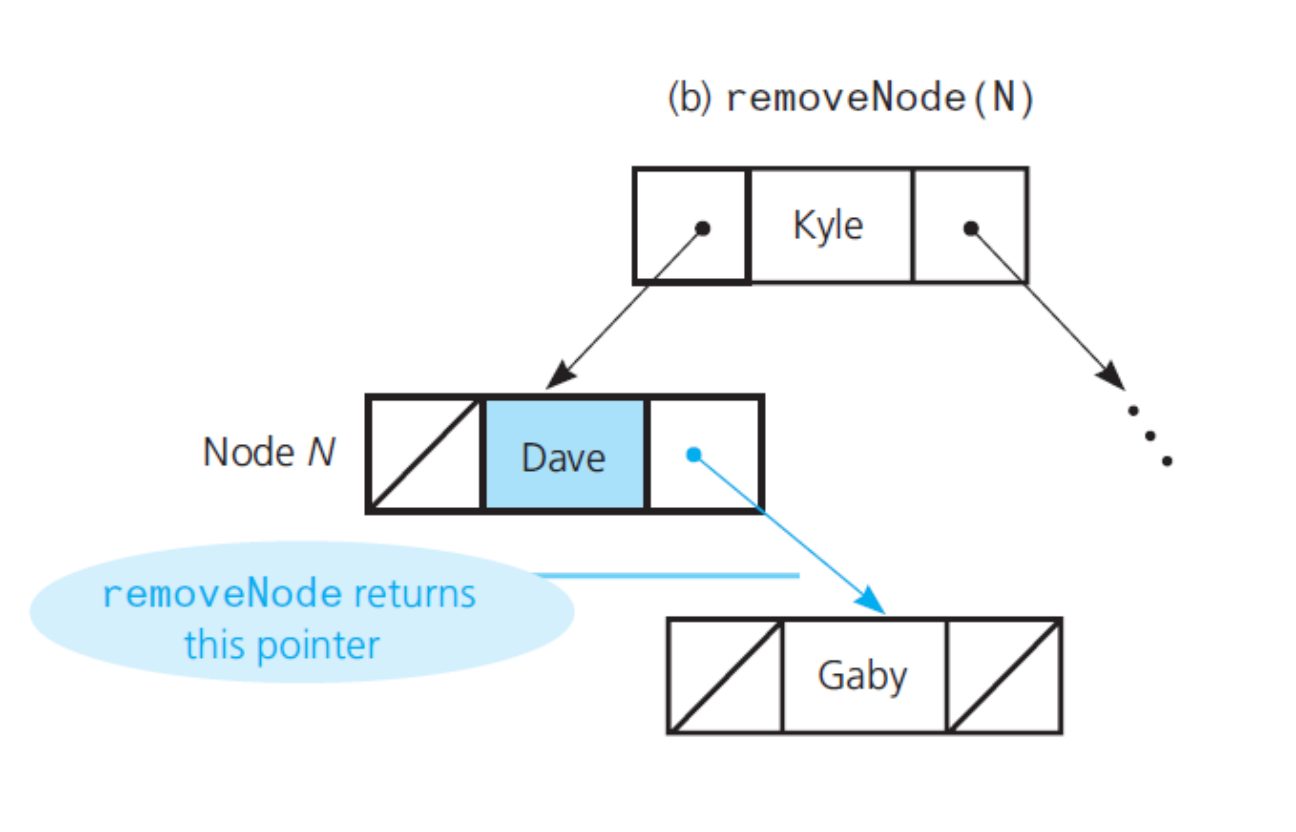


FIGURE 16-14 Recursive removal of node N

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# Link-Based Implementation of the ADT Binary Search Tree

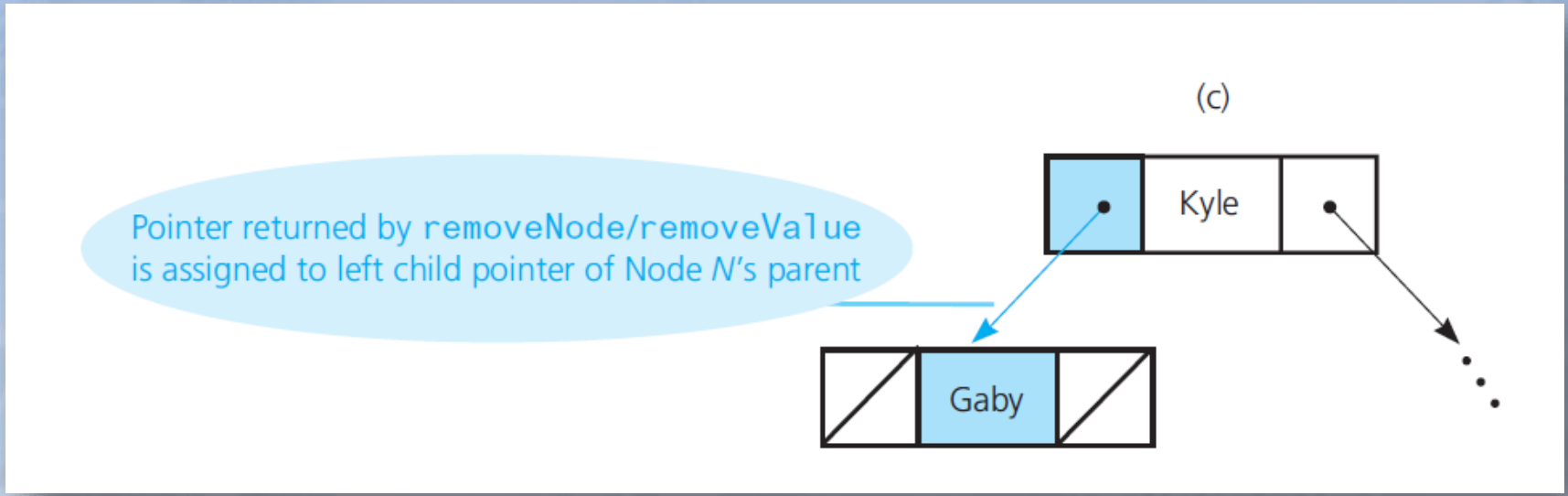


FIGURE 16-14 Recursive removal of node N

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# Link-Based Implementation of the ADT Binary Search Tree

```
// Locates the node in the binary search tree to which subTreePtr points and that contains
// the value target. Returns either a pointer to the located node or nullptr if such a
// node is not found.
findNode(subTreePtr: BinaryNodePointer, target: ItemType): BinaryNodePointer
{
   if (subTreePtr == nullptr)
      return nullptr                                  // Not found

   else if (subTreePtr->getItem() == target)
      return subTreePtr;                              // Found
   else if (subTreePtr->getItem() > target)
      // Search left subtree
      return findNode(subTreePtr->getLeftChildPtr(), target)
   else
      // Search right subtree
      return findNode(subTreePtr->getRightChildPtr(), target)
}
```

Algorithm for findNode

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# The Class BinarySearchTree

```
/** Link-based implementation of the ADT binary search tree.
 @file BinarySearchTree.h */

#ifndef BINARY_SEARCH_TREE_
#define BINARY_SEARCH_TREE_

#include "BinaryTreeInterface.h"
#include "BinaryNode.h"
#include "BinaryNodeTree.h"
#include "NotFoundException.h"
#include "PrecondViolatedExcept.h"
#include <memory>

template<class ItemType>
class BinarySearchTree : public BinaryNodeTree<ItemType>
{
private:
   std::shared_ptr<BinaryNode<ItemType>> rootPtr;
```

LISTING 16-4 A header file for the link-based implementation of the class BinarySearchTree

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# The Class BinarySearchTree

```
protected:
   //-----------------------------------------------------------
   //    Protected Utility Methods Section:
   //    Recursive helper methods for the public methods.
   //-----------------------------------------------------------
   // Places a given new node at its proper position in this binary
   // search tree
   auto placeNode(std::shared_ptr<BinaryNode<ItemType>> subTreePtr,
                  std::shared_ptr<BinaryNode<ItemType>> newNode);

   // Removes the given target value from the tree while maintaining a
   // binary search tree.
   auto removeValue(std::shared_ptr<BinaryNode<ItemType>> subTreePtr,
                                    const ItemType target,
                                    bool& isSuccessful) override;

   // Removes a given node from a tree while maintaining a binary search tree.
   auto removeNode(std::shared_ptr<BinaryNode<ItemType>> nodePtr);
```

LISTING 16-4 A header file for the link-based implementation of the class BinarySearchTree

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# The Class BinarySearchTree

```
    // Removes a given node from a tree while maintaining a binary search tree.
    auto removeNode(std::shared_ptr<BinaryNode<ItemType>> nodePtr);

    // Removes the leftmost node in the left subtree of the node
    // pointed to by nodePtr.
    // Sets inorderSuccessor to the value in this node.
    // Returns a pointer to the revised subtree.
    auto removeLeftmostNode(std::shared_ptr<BinaryNode<ItemType>>subTreePtr,
                                          ItemType& inorderSuccessor);

    // Returns a pointer to the node containing the given value,
    // or nullptr if not found.
    auto findNode(std::shared_ptr<BinaryNode<ItemType>> treePtr,
                                 const ItemType& target) const;

public:
    //------------------------------------------------------------
    //   Constructor and Destructor Section.
    //------------------------------------------------------------
    BinarySearchTree();
    BinarySearchTree(const ItemType& rootItem);
    BinarySearchTree(const BinarySearchTree<ItemType>& tree);
    virtual ~BinarySearchTree();
```

LISTING 16-4 A header file for the link-based implementation of the class BinarySearchTree

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# The Class BinarySearchTree

```
    BinarySearchTree(const ItemType& rootItem);
    BinarySearchTree(const BinarySearchTree<ItemType>& tree);
    virtual ~BinarySearchTree();

    //-----------------------------------------------------------
    //   Public Methods Section.
    //-----------------------------------------------------------
    bool isEmpty() const;
    int getHeight() const;
    int getNumberOfNodes() const;
    ItemType getRootData() const throw(PrecondViolatedExcept);
    void setRootData(const ItemType& newData);
    bool add(const ItemType& newEntry);
    bool remove(const ItemType& target);
    void clear();
    ItemType getEntry(const ItemType& anEntry) const throw(NotFoundException);
```

LISTING 16-4 A header file for the link-based implementation of the class BinarySearchTree

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# The Class BinarySearchTree

```
    bool contains(const ItemType& anEntry) const;

    //-----------------------------------------------------------
    //  Public Traversals Section.
    //-----------------------------------------------------------
    void preorderTraverse(void visit(ItemType&)) const;
    void inorderTraverse(void visit(ItemType&)) const;
    void postorderTraverse(void visit(ItemType&)) const;

    //-----------------------------------------------------------
    //   Overloaded Operator Section.
    //-----------------------------------------------------------
    BinarySearchTree<ItemType>&
                operator=(const BinarySearchTree<ItemType>& rightHandSide);
};  // end BinarySearchTree
#include "BinarySearchTree.cpp"
#endif
```

LISTING 16-4 A header file for the link-based implementation of the class BinarySearchTree

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# Link-Based Implementation of the ADT Binary Search Tree

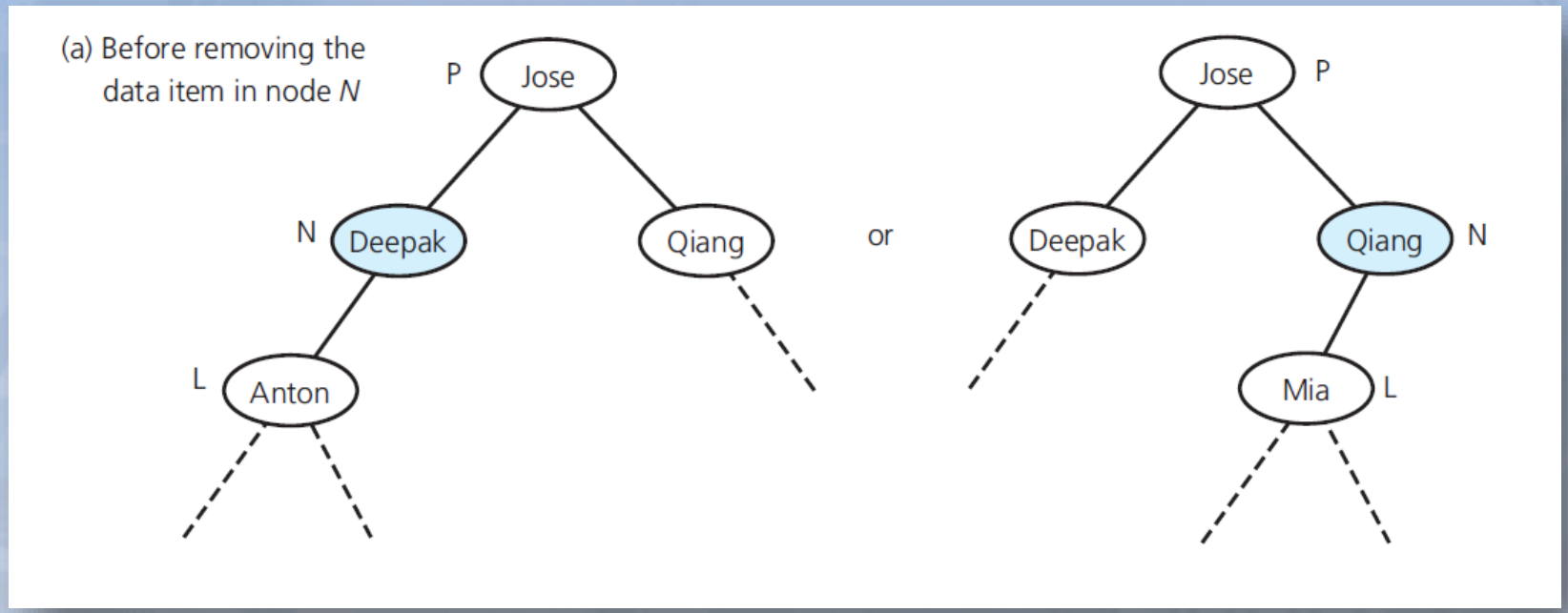


Figure 16-9 Case 2 for removeValue: The data item to remove is in a node N that has only a left child and whose parent is node P

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# Link-Based Implementation of the ADT Binary Search Tree

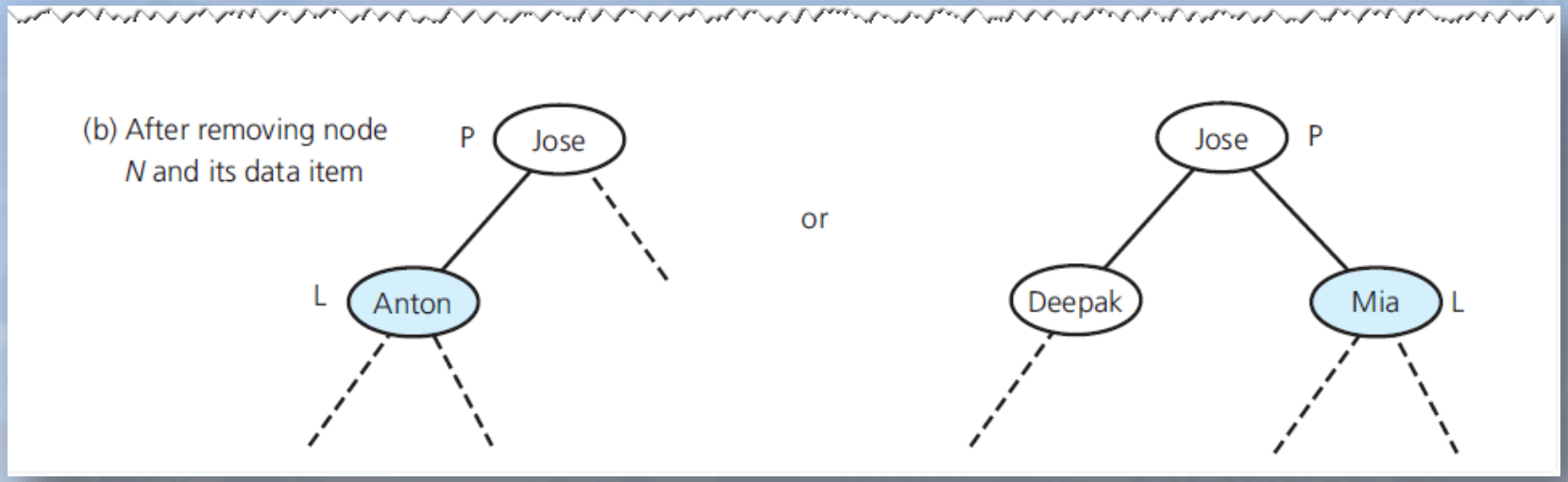


FIGURE 16-9 Case 2 for removeValue: The data item to remove is in a node *N* that has only a left child and whose parent is node *P*

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# Saving a Binary Search Tree in a File

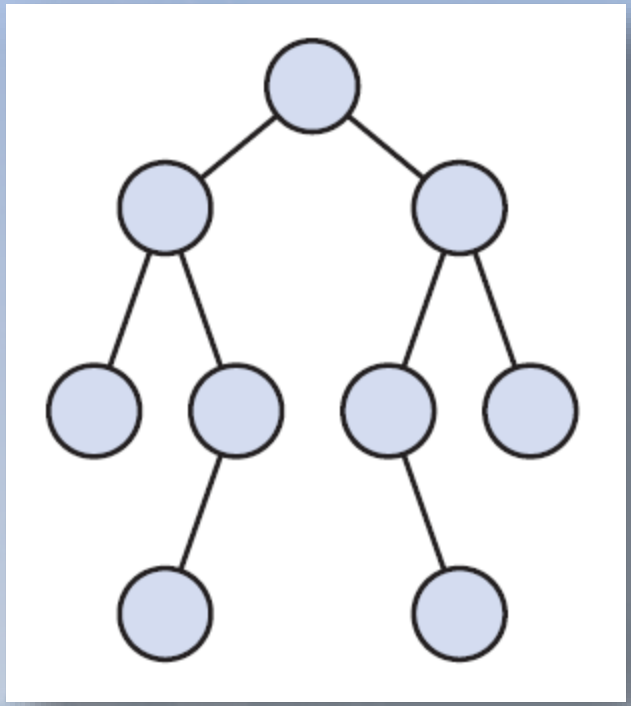

FIGURE 16-17 A tree of minimum height that is not complete

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